

ESIL', Kazakhstan

WMO: 350680

Lat: 51.9733N

Lon: 66.4183E

Elev: 221

StdP: 98.70

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-29.3	-35.2	0.1	-32.0	-32.3	0.2	-29.2	12.1	-9.2	10.0	-10.6	1.1	50	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.8	18.0	31.0	17.6	29.2	17.2	19.9	28.1	19.1	27.6	18.4	26.6	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	12.7	22.7	16.3	11.9	22.1	15.3	11.2	21.5	58.1	28.0	55.3	27.6	52.9	26.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.7	9.4	8.2	DB	-35.4	35.9	2.5	2.1	-37.2	37.4	-38.7	38.6	-40.1	39.8	-41.9	41.3	(4)
(5)			WB	-35.4	21.6	2.4	1.2	-37.1	22.5	-38.5	23.2	-39.9	23.9	-41.6	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-16.2	-14.6	-6.8	6.0	14.0	19.0	20.1	18.9	12.6	4.8	-5.3	-12.8		
	DBStd	14.51	7.33	6.83	6.58	6.14	4.84	4.25	3.55	4.37	4.80	5.06	7.46	7.29		
	HDD10.0	3555	813	688	521	147	18	1	0	1	27	173	459	708		
	HDD18.3	5676	1072	922	779	371	147	42	23	46	181	418	709	967		
	CDD10.0	1147	0	0	0	26	143	270	313	278	104	13	0	0		
	CDD18.3	227	0	0	0	0	13	62	78	65	9	0	0	0		
	CDH23.3	2927	0	0	0	11	244	804	904	817	143	3	0	0		
	CDH26.7	1053	0	0	0	1	60	302	322	322	46	0	0	0		
(14) Wind	WSAvg	3.1	2.8	3.2	3.4	3.4	3.4	3.1	3.0	2.7	2.8	3.0	3.1	3.3		
(15) Precipitation	PrecAvg	291	14	13	16	19	33	36	52	28	16	21	22	21		
	PrecMax	465	34	31	39	43	80	137	104	71	34	59	38	49		
	PrecMin	154	2	1	1	1	3	6	9	3	2	1	7	4		
	PrecStd	78	8	7	10	11	20	30	27	20	9	12	9	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.8	10.3	24.9	31.0	34.2	34.8	35.0	31.6	23.0	10.2	1.2		
		MCWB	0.1	-0.1	6.3	13.2	16.1	18.4	18.6	18.4	16.7	11.3	6.6	0.0		
	2%	DB	-1.1	-0.5	5.1	21.5	28.2	32.1	32.2	32.5	27.2	19.1	7.2	0.1		
		MCWB	-1.7	-1.1	3.1	11.7	15.3	17.6	18.4	18.0	14.3	10.1	5.1	-0.6		
	5%	DB	-3.0	-2.1	3.1	18.8	26.0	30.3	30.0	30.3	24.1	16.0	4.9	-1.3		
		MCWB	-3.6	-2.7	1.6	10.8	14.4	17.1	18.4	17.7	13.6	8.7	3.2	-2.0		
	10%	DB	-4.9	-4.1	1.6	16.0	23.9	28.0	28.2	27.9	21.4	13.1	3.1	-3.1		
		MCWB	-5.5	-4.7	0.6	9.7	13.5	16.7	17.8	17.0	12.5	7.6	1.7	-3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.1	6.3	14.6	17.7	20.4	21.2	20.4	17.4	12.7	7.7	0.5		
		MCDB	0.5	0.7	9.8	21.0	26.4	29.1	28.5	29.4	28.0	20.7	9.0	1.0		
	2%	WB	-1.6	-1.1	3.2	12.9	16.3	19.3	20.3	19.3	15.4	10.9	5.2	-0.5		
		MCDB	-1.2	-0.5	4.8	19.6	25.4	28.1	27.8	28.8	24.4	17.2	7.1	0.0		
	5%	WB	-3.5	-2.7	1.7	11.4	15.3	18.4	19.4	18.5	14.2	9.4	3.4	-2.0		
		MCDB	-3.0	-2.2	2.8	17.6	23.8	27.3	27.1	27.7	22.1	14.7	4.8	-1.3		
	10%	WB	-5.4	-4.7	0.7	9.9	14.3	17.5	18.7	17.7	13.1	8.0	1.8	-3.8		
		MCDB	-4.9	-4.1	1.7	15.8	22.3	26.1	26.2	26.2	20.5	12.6	3.0	-3.1		
(35) Mean Daily Temperature Range	MDBR		8.2	8.8	8.6	11.1	13.2	12.7	12.1	12.9	12.5	10.0	7.2	7.9		
	5% DB	MCDBR	7.5	6.7	7.5	16.1	16.9	16.1	15.4	16.2	16.5	15.9	8.2	7.1		
		MCWBR	7.4	6.5	5.9	8.2	7.2	6.2	5.6	6.0	7.3	8.7	6.4	6.8		
	5% WB	MCDBR	7.5	6.7	7.2	14.4	14.4	13.6	12.6	14.3	14.4	13.3	7.4	6.9		
		MCWBR	7.4	6.6	5.9	8.2	7.1	6.1	5.4	6.0	6.9	7.9	6.2	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.325	0.378	0.411	0.408	0.407	0.421	0.420	0.367	0.349	0.327	0.293		
	taud		2.060	2.008	1.961	2.118	2.212	2.287	2.248	2.225	2.324	2.277	2.149	2.057		
	Ebn at Noon		670	758	789	819	844	848	827	804	810	739	625	591		
	Edh at Noon		79	112	144	140	134	125	128	125	100	86	73	67		
(44) All-Sky Solar Radiation	RadAvg		1.33	2.46	3.86	4.77	5.95	6.30	5.88	5.05	3.72	2.03	0.95	0.88		
	RadStd		0.10	0.14	0.41	0.48	0.46	0.47	0.42	0.43	0.28	0.27	0.10	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page